

10.7.71

EE25BTECH11022 - sankeerthan

Question: The area formed by the positive X axis and the normal and the tangent to the circle $x^2 + y^2 = 4$ at $(1, \sqrt{3})$ is _____.

Solution: Equation of the circle

$$\mathbf{X}^T \mathbf{X} = 4, \text{ where } \mathbf{X} = \begin{pmatrix} x \\ y \end{pmatrix} \quad (1)$$

Given point of contact

$$\mathbf{P} = \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix} \quad (2)$$

Equation of the tangent at $\mathbf{P}$

$$\mathbf{P}^T \mathbf{X} = 4 \quad (3)$$

Let intersection of tangent with X-axis be $\mathbf{A} = \begin{pmatrix} a \\ 0 \end{pmatrix}$

$$\mathbf{P}^T \mathbf{A} = 4 \quad (4)$$

$$\begin{pmatrix} 1 & \sqrt{3} \end{pmatrix} \begin{pmatrix} a \\ 0 \end{pmatrix} = 4 \implies a = 4 \quad (5)$$

$$\mathbf{A} = \begin{pmatrix} 4 \\ 0 \end{pmatrix} \quad (6)$$

Equation of the normal at $\mathbf{P}$

$$\mathbf{X} = t \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix}, t \in \mathbb{R} \quad (7)$$

Let intersection of normal with X-axis be $\mathbf{N} = \begin{pmatrix} x \\ 0 \end{pmatrix}$

$$\mathbf{N} = t \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix} \implies 0 = t \sqrt{3} \implies t = 0 \implies \mathbf{N} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad (8)$$

Area of triangle formed by points $\mathbf{O}, \mathbf{A}, \mathbf{P}$

$$\mathbf{O} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \mathbf{A} = \begin{pmatrix} 4 \\ 0 \end{pmatrix}, \mathbf{P} = \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix} \quad (9)$$

$$S = \frac{1}{2} \left| \begin{pmatrix} 0 & 0 & 1 \\ 4 & 0 & 1 \\ 1 & \sqrt{3} & 1 \end{pmatrix} \right| \quad (10)$$

$$= \frac{1}{2} |0(0 - \sqrt{3}) - 0(4 - 1) + 1(4\sqrt{3} - 0)| \quad (11)$$

$$= \frac{1}{2} |4\sqrt{3}| = 2\sqrt{3} \quad (12)$$

Therefore, required area formed $= 2\sqrt{3}$

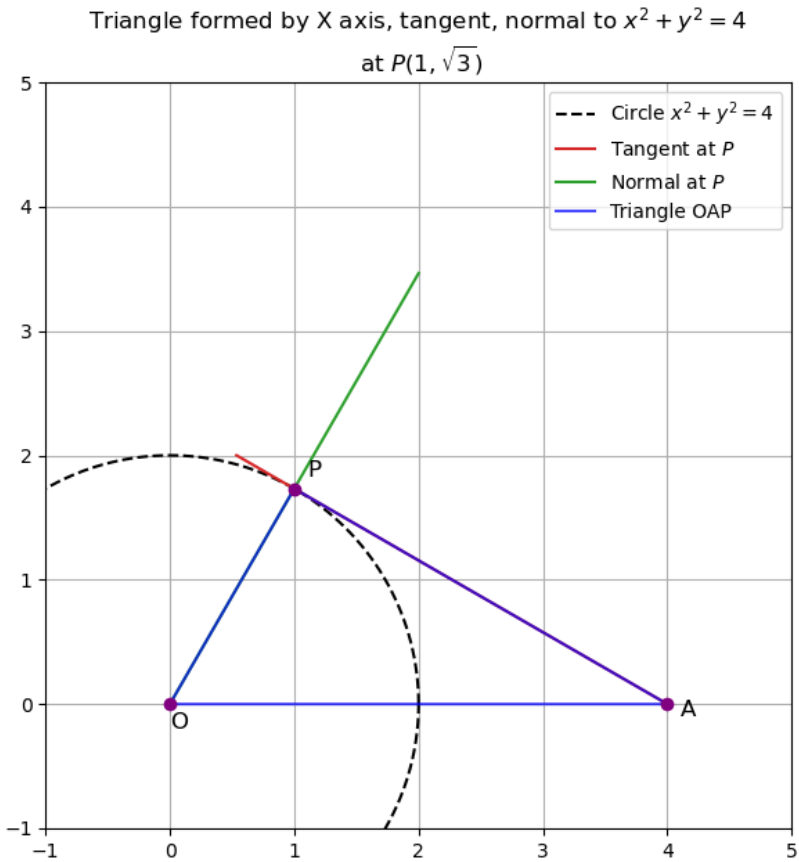


Fig. 0.1